

AIMLCZG557

Slide Exercise & Practice Questions

Topic-wise question bank - questions only

Compiled from all distinct lecture and webinar slide sets in the local ACI workspace

Original slide questions and diagrams are preserved. Duplicate deck copies are consolidated.

24 June 2026

Contents

1

Intelligent Agents, PEAS & Task Environments

CS02 exercises 1-3

2

Problem Formulation

CS03 exercise 1

3

Informed Search, A* & Heuristic Design

CS03, CS04, CS06 and webinar questions

4

Local Search & Hill Climbing

CS05 and webinar practice

5

Genetic Algorithms

CS05/CS06 duplicate exercise consolidated

6

Adversarial Search, Minimax & Evaluation

CS07 exercises 1-2



Intelligent Agents, PEAS & Task Environments

Exercises on PEAS specification and environment classification.

Exercise 1: Write the PEAS for an Interactive English tutor such as Grammarly

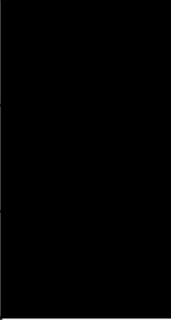
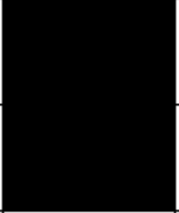


- Performance measure: ?
- Environment: ?
- Actuators: ?
- Sensors: ?

Exercise 2: Path Finding Robot



PEAS for this problem?

1	2	3	4	5	6
	8		10	11	12
13	14		16	17	18
19	20		22	23	24
25	26	27			30
	32	33		35	36
37	38	39	40	41	42

Observability?

Number of Agents?

Determinism?

Dynamicity?

Episodic or Sequential?

No of States?

Exercise 3: Identify the Environment



- 15-Puzzle game
- Poker
- Medical Diagnosis
- Taxi Driving



2

Problem Formulation

Formulate initial state, actions, transition model, goal test and path cost.

Exercise 1



For the below tasks:

1. Planning a trip with a delivery drone
2. Solving a maze
3. Scheduling patient appointments in a clinic

Formulae **the problem** by specifying:

- Initial state
- Possible actions
- Transition model (effects of actions)
- Goal test
- Path cost (optional)

Think about:

- Any **missing components**
- Assess **well-formulated-ness** (Does it meet criteria for a well-formed problem?)

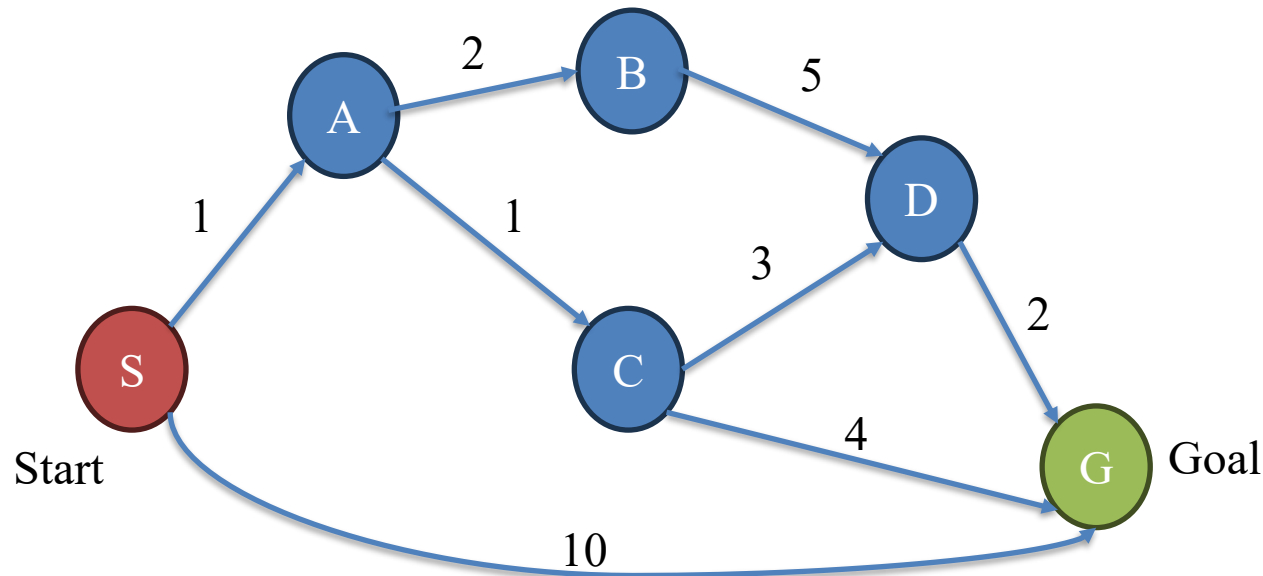


3

Informed Search, A* & Heuristic Design

A*, GBFS, admissibility, consistency, relaxed problems and exam-style grid searches.

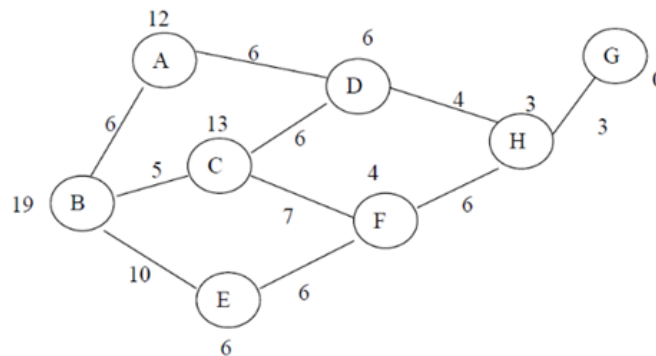
Exercise 2



Apply A* algorithm on the above graph, given the heuristics:
 $h(S)=5$, $h(A)=3$, $h(B)=4$, $h(C)=2$, $h(D)=6$ and $h(G) = 0$

Exercise 1

- Consider the following weighted graph. The number beside each node represents its heuristic value $h(n)$, which estimates the cost from that node to the goal node G .
- The start node is B , and the goal node is G .



- Apply GBFS on this graph.
- Apply A* Algorithm on this graph.
- Compare your findings from the above two
- Check if the heuristics is admissible and consistent.

Exercise 2

- A warehouse robot has to collect three packages and reach the exit.
- The warehouse has rooms connected by corridors. Some corridors have locked doors. The robot can move from one room to an adjacent room with cost 1. To pass through a locked door, the robot must first collect the matching key.
- **Warehouse layout** Rooms:

$S, A, B, C, D, E, F, G, X$

where:

- S = start room
- X = exit room
- Packages are located at C , F , and G
- Key K_1 is located at B
- Key K_2 is located at E

The robot must collect all three packages at C , F , and G , then reach X .

Exercise 2 Continued

Corridor	Cost	Condition
(S-A)	1	Open
(A-B)	1	Open
(A-C)	2	Open
(B-D)	2	Requires (K1)
(C-D)	2	Open
(D-E)	1	Open
(E-F)	2	Requires (K2)
(D-G)	3	Open
(G-X)	2	Open
(F-X)	2	Open

Part A: State Representation

Define a state representation for this problem.

Your state should include:

1. Current room of the robot.
2. Packages already collected.
3. Keys already collected.

Write the initial state and one possible goal state.

Exercise 2 Continued

Part B: Relaxed Problems

A relaxed problem is obtained by removing one or more constraints from the original problem. Consider the following relaxed versions of the warehouse problem:

- **Relaxation 1:** Ignore all locked-door constraints. The robot can pass through any corridor.
- **Relaxation 2:** Ignore package collection order. Estimate the cost only as the shortest distance from the current room to the exit.
- **Relaxation 3:** Assume the robot can collect any remaining package instantly once it enters the same connected component, but it must still move to the exit.
- **Relaxation 4:** Ignore the robot's current location and estimate only the minimum cost needed to connect all uncollected package locations and the exit.

Answer the following:

1. For each relaxed problem, explain which constraint has been removed.
2. Which relaxed problem is likely to give the weakest heuristic?
3. Which relaxed problem is likely to give the strongest heuristic?
4. Why does the solution cost of a relaxed problem give an admissible heuristic for the original problem?
5. Construct few heuristics by picking some relaxed version of the problem.

Exercise 1

A warehouse robot begins at C1, tasked with collecting Items (It) and reaching the Dock (DV) at C4 on a 4x4 grid. The robot must navigate hazards like blocked cells (#), Spills (SP) with high movement costs, while leveraging Energy packs (E) to reduce costs. Using the A* algorithm, the robot calculates the optimal path by considering both actual path cost and a heuristic. The agent has four distinct actions: Move Up, Move Down, Move Left, and Move Right. However, it cannot move into blocked cells. The warehouse is represented as a 4×4 grid below:

	1	2	3	4
A	It	E		SP
B	SP	It	E	SP
C	S	E	It	DV
D	It	SP	SP	E

Entering SP: +12; collecting It: −8 (reward); energy pack E: +1.
Heuristic $h(n) = \text{Manhattan}(\text{robot}, \text{DV}) + (\# \text{ remaining Items}) + (\# \text{ adjacent SP to robot in resultant state})$.

- Expand and depict the search tree for exactly 2 levels. (Level 0 = Start).
- Calculate Path costs and heuristics (g/h/f) for all generated nodes.
- Apply A* search and show all the steps.



Question 1:

A Rescue Agent is deployed in a flood-affected city to deliver medicines from the base camp at (X1, Y1) to a relief shelter located at (X5, Y4). The agent must efficiently plan the route to ensure the timely delivery of the medical supplies. The city is represented as a 5x5 grid. The agent is equipped with four distinct actions: Move Up, Move Down, Move Left, and Move Right. However, it cannot move into waterlogged and blocked cells. The city is represented as a 5x5 grid below:



	Y1	Y2	Y3	Y4	Y5
X1	S	N	W	N	N
X2	N	W	N	N	B
X3	N	N	W	E	N
X4	B	N	N	W	N
X5	N	N	E	G	N

Legend:

S = Start (Base Camp)

G = Goal (Relief Shelter)

N = Normal cell (cost +1)

W = Waterlogged (+5 cost)

E = Energy pack (-2 cost)

B = Blocked (not traversable)



Heuristic:

$H(n) = \text{Manhattan Distance to Goal} +$
No. of Waterlogged cells adjacent to position

For the above problem do the following:

- a) Expand and depict the search tree for exactly 3 levels. (Level 0 = Start).
- b) Calculate path cost and heuristic values for all nodes at Level 1–3.
- c) Apply A* search and show all steps.

Manhattan distance = For two points $(P(x_1, y_1))$ and $(Q(x_2, y_2))$, the formula is:
 $D = |x_1 - x_2| + |y_1 - y_2|$

PEAS and Environment example



Question 2:

A large supermarket chain wants to implement an Autonomous Checkout System where customers can pick items and walk out without scanning them manually. The system uses cameras, weight sensors, and RFID scanners to track selected items, update the bill, and charge customers automatically through a mobile app.

- a. Provide the complete problem formulation.
- b. Provide the PEAS description.
- c. Identify the various dimensions of the task environment with appropriate justification for each.



4

Local Search & Hill Climbing

Local beam selection and extension tasks for hill-climbing code.

Exercise 1

- Suppose local beam search has $k = 3$ and the following current states:

Current State	Heuristic Value
S1	12
S2	15
S3	10

- Their successors are:

Successor	Heuristic Value
A	16
B	11
C	14
D	18
E	13
F	17

If the goal is to maximize, which three states will be selected for the next beam?

Hill Climbing - Food for Thought

- Add code to convert stochastic hill climbing into random-restart hill climbing.
- Solve the travel-optimization problem using simulated annealing.
- Implement the solution for the same problem using a genetic algorithm.



5

Genetic Algorithms

Single-point crossover and fitness-based selection.

Exercise 2

1. Given two parent chromosomes:

➤ Parent 1: [1, 2, 3, 4, 5, 6, 7, 8]

Parent 2: [8, 7, 6, 5, 4, 3, 2, 1]

➤ Using single-point crossover after the 4th position, generate two children.

2. Suppose the fitness values of four individuals are:

Individual	Fitness
A	10
B	30
C	40
D	20

Which individual has the highest chance of being selected? Why?



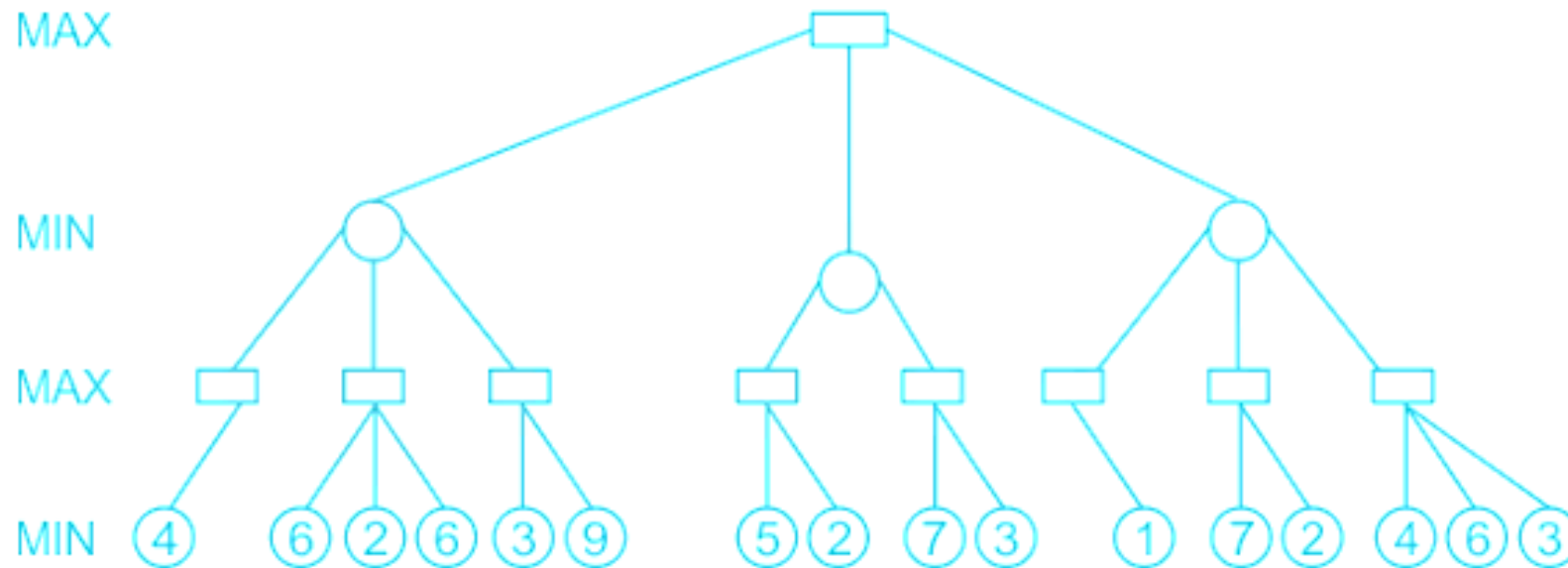
6

Adversarial Search, Minimax & Static Evaluation

Minimax tree backup and a two-ply castle-battle evaluation problem.

Exercise 1

Consider the following minimax game tree search



What will be the value propagated at the root?

Exercise 2: Castle Battle Game

- Two players are playing a small strategy game.
- **MAX** is the attacking army.
- **MIN** is the defending army.
- MAX wants to capture the castle.
- MIN wants to prevent MAX from capturing the castle.
- The game is not fully played until the end. We will search only **2 moves ahead** and then use a **static evaluation function**.

Part A: Draw the Game Tree

At the current position, MAX has three possible moves:

1. **Attack Left Gate**
2. **Attack Main Gate**
3. **Attack Right Gate**

After each MAX move, MIN has two possible replies.

MAX move	MIN reply 1	MIN reply 2
Attack Left Gate	Reinforce Left Gate	Counterattack Left
Attack Main Gate	Reinforce Main Gate	Block Supply Route
Attack Right Gate	Reinforce Right Gate	Counterattack Right

Exercise 2: Castle Battle Game

Task 1

- Draw the game tree up to depth 2.
- The root node should be MAX's turn.
- The second level should be MIN's turn.
- The leaf nodes should be the board positions after MIN replies.

Part B: Design a Static Evaluation Function

At each leaf node, the game is not over. So, we need to estimate how good the position is for MAX.

Assume we judge the position using four features:

Feature	Meaning
Soldier advantage	MAX soldiers – MIN soldiers
Gate damage	Damage done to castle gates
MAX safety	How safe MAX's army is
Supply control	Whether MAX controls the supply route

$$\text{Evaluation} = 2 \times \text{Soldier Advantage} + 3 \times \text{Gate Damage} + 2 \times \text{MAX Safety} + 4 \times \text{Supply Control}$$

Exercise 2: Castle Battle Game

Leaf	Soldier Advantage	Gate Damage	MAX Safety	Supply Control
L1	2	3	2	0
L2	1	4	0	1
M1	3	2	3	0
M2	1	3	1	2
R1	2	2	2	1
R2	0	5	0	1

Given the above feature values (leaf), compute the evaluation scores using the eval function.